

SQL Interview Paper (Intermediate to Advanced).

Section A – Ranking & Aggregation.

1. Write an SQL query to fetch the **second highest salary** from the **Employee** table. That's a great format for SQL interview preparation. Here's a professional template for your first question.

Question 1: Find the Second Highest Salary

Problem Statement

You are given an **Employee** table containing employee details, including their salaries.

Write an SQL query to fetch the employee(s) who have the second highest salary.

Note: If multiple employees have the same second highest salary, return all of them.

Table Structure

Employee

Column Name	Data Type	Description
EmpID	INT	Unique Employee ID
EmpName	VARCHAR(100)	Employee Name
DepartmentID	INT	Department ID
Salary	DECIMAL(10,2)	Employee Salary

Sample Data

EmpID	EmpName	DepartmentID	Salary
101	John	1	90000
102	Alice	2	120000
103	Bob	1	110000
104	David	2	120000
105	Emma	3	95000
106	Chris	1	105000

Expected Output

EmpID	EmpName	Salary
103	Bob	110000

Solution

There are multiple ways to solve this problem:

1. Using `DENSE_RANK()` (Recommended)
2. Using `MAX()`
3. Using a Subquery
4. Using `LIMIT/OFFSET` (Database-specific)

The best approach is to use the **`DENSE_RANK()`** window function because it correctly handles duplicate salaries.

SQL Query (Using `DENSE_RANK`)

```
WITH SalaryRank AS
(
    SELECT *,
           DENSE_RANK() OVER(ORDER BY Salary DESC) AS SalaryRank
    FROM Employee
)
SELECT EmpID,
       EmpName,
```

```
Salary
FROM SalaryRank
WHERE SalaryRank = 2;
```

Alternative Solution (Using MAX)

```
SELECT *
FROM Employee
WHERE Salary =
(
    SELECT MAX(Salary)
    FROM Employee
    WHERE Salary <
    (
        SELECT MAX(Salary)
        FROM Employee
    )
);
```

Tips

- Use **DENSE_RANK()** when duplicate salaries should share the same rank.
- Avoid using **LIMIT** or **TOP** in interview answers unless the interviewer specifies a particular database.
- The **MAX()** approach works well but becomes difficult for the Nth highest salary.
- If two employees earn the second highest salary, both should be returned.
- Mention the time complexity and readability of your chosen approach during interviews.
- Be prepared for follow-up questions such as:
 - Find the 3rd highest salary.
 - Find the Nth highest salary.
 - Find the second highest salary department-wise.
 - Explain the difference between **RANK()**, **DENSE_RANK()**, and **ROW_NUMBER()**.

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1. Write an SQL query to fetch the **second highest salary in each department**.
 2. Write an SQL query to find the **top-paid employee in each department**.
 3. Write an SQL query to find the **top 3 highest-paid employees in each department**.

4. Write an SQL query to find employees whose salary is **higher than the average salary of their department**.
5. Write an SQL query to find employees whose salary is **lower than the average salary of their department**.
6. Write an SQL query to calculate the **average salary for each department**.
7. Write an SQL query to find the department having the **highest average salary**.

Section B – Joins & Self Joins.

9. Write an SQL query to find employees who earn **more than their manager**.
10. Write an SQL query to display each employee along with their manager's name.
11. Write an SQL query to find employees who **do not have a manager**.
12. Write an SQL query to find managers who have **more than five employees reporting to them**.

Section C – Window Functions.

13. Write an SQL query to assign **Rank, Dense Rank, and Row Number** based on salary within each department.
14. Write an SQL query to fetch the **third highest salary** in each department.
15. Write an SQL query to identify employees having **duplicate salaries** within the same department.

Section D – Group By & Having.

16. Write an SQL query to find departments having **more than 10 employees**.
17. Write an SQL query to find departments where the **average salary exceeds ₹100,000**.
18. Write an SQL query to count the number of employees in each department.

Section E – Advanced SQL

19. Write an SQL query to find employees whose salary is **greater than the salary of all employees in the Sales department**.
20. Write an SQL query to find the employee(s) with the **highest salary without using MAX()**.
21. Write an SQL query to fetch the **Nth highest salary**, where **N** is provided as an input parameter.
22. Write an SQL query to find employees who have the **same salary as another employee**.
23. Write an SQL query to find employees whose salary is **between the department average salary and the department maximum salary**.
24. Write an SQL query to find the **department-wise salary difference** between the highest-paid and lowest-paid employee.
25. Write an SQL query to identify departments where **all employees earn more than ₹50,000**.
26. Write an SQL query to display the **salary percentage contribution** of each employee within their department.
27. Write an SQL query to find employees who joined **before their manager**.
28. Write an SQL query to find employees who share the **same manager and the same salary**.

29. Write an SQL query to find employees who have the **longest tenure** in each department.
30. Write an SQL query to display the **salary trend** by assigning salary bands (Low, Medium, High) using a **CASE** statement.

Bonus Scenario-Based Questions.

1. Find customers who have never placed an order.
2. Find products that were never sold.
3. Find duplicate records from a table.
4. Delete duplicate records while keeping the latest record.
5. Find consecutive duplicate values.
6. Find the latest record for each employee.
7. Find employees who joined in the last 30 days.
8. Find departments without any employees.
9. Swap values of two columns without using a temporary variable.
10. Find the second most recent order for each customer.
11. Calculate a running total of salary by department.
12. Find missing IDs in a sequence.
13. Pivot employee salaries department-wise.
14. Unpivot department columns into rows.
15. Find the employee with the highest salary in the company using a Common Table Expression (CTE).

